

Physical Chemistry (I)

Lecture 02

Basic Concepts of Thermodynamics

Perfect Gas



Focus vs. Chapter

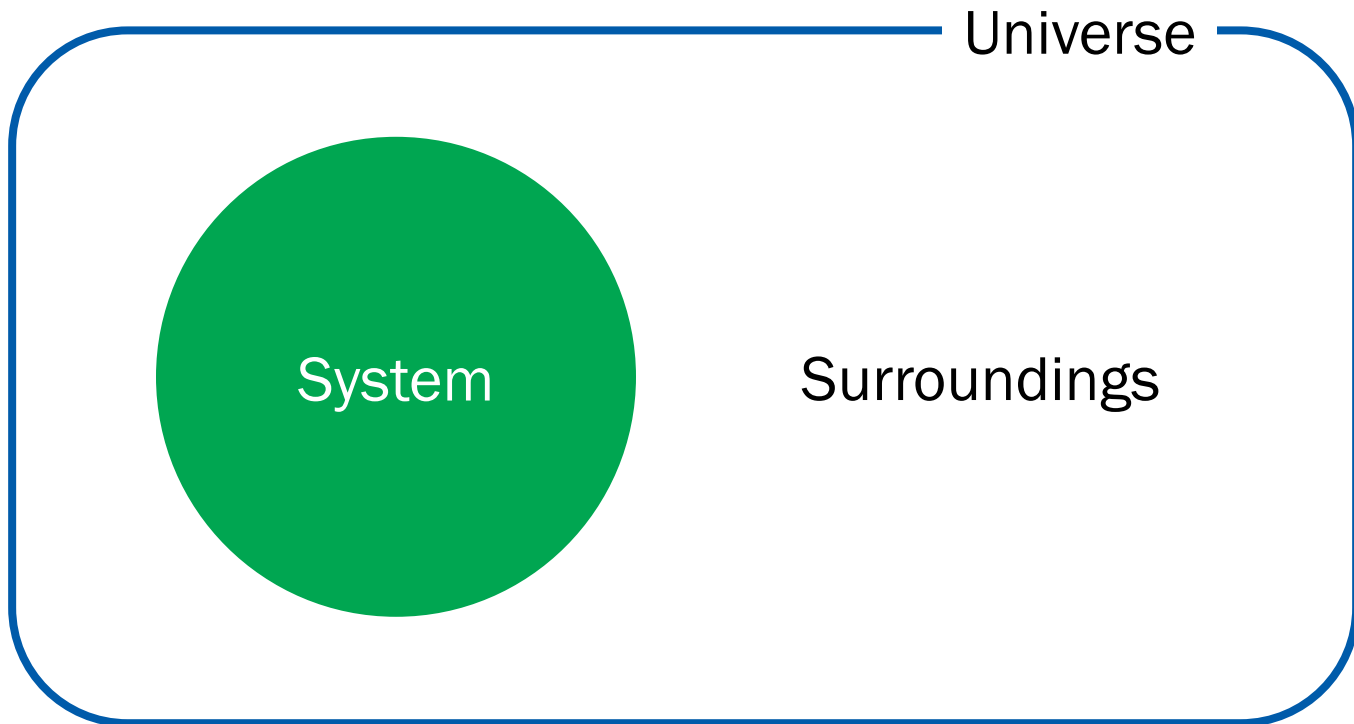
Atkins	This course
Focus 1	Chapter 2
Focus 2	Chapter 1, Chapter 3
Focus 3	Chapter 4, Chapter 5
Focus 4	Chapter 6
Focus 5	Chapter 7
Focus 6	Chapter 8, Chapter 9

Thermodynamics

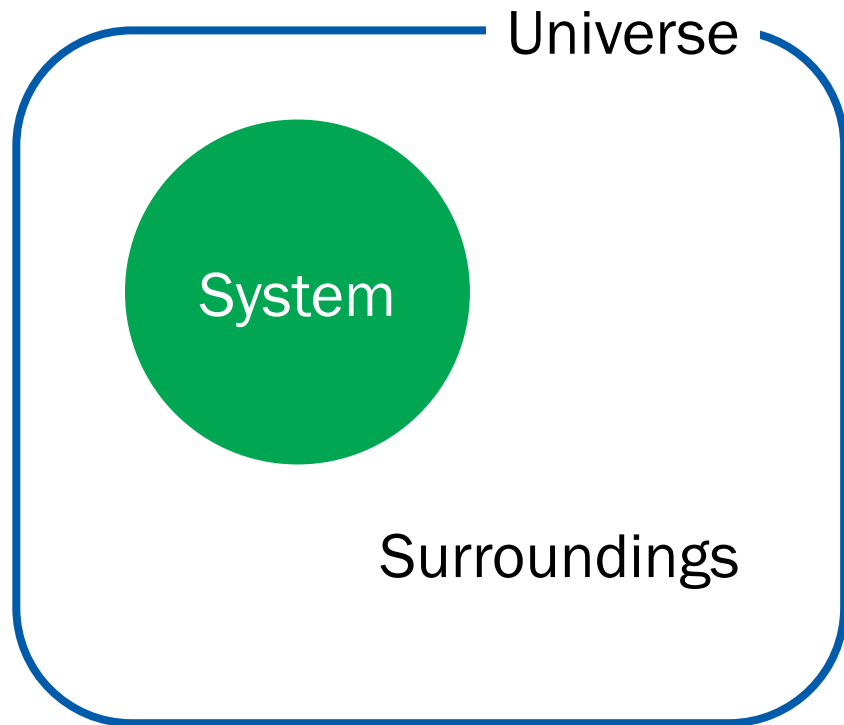
- Thermodynamics
 - thermo: “heat” (θέρμη)
 - dynamics: “power” (δύναμις)
- Study of ...
 - heat and temperature
 - and their relation to energy and work

1. Basic Concepts

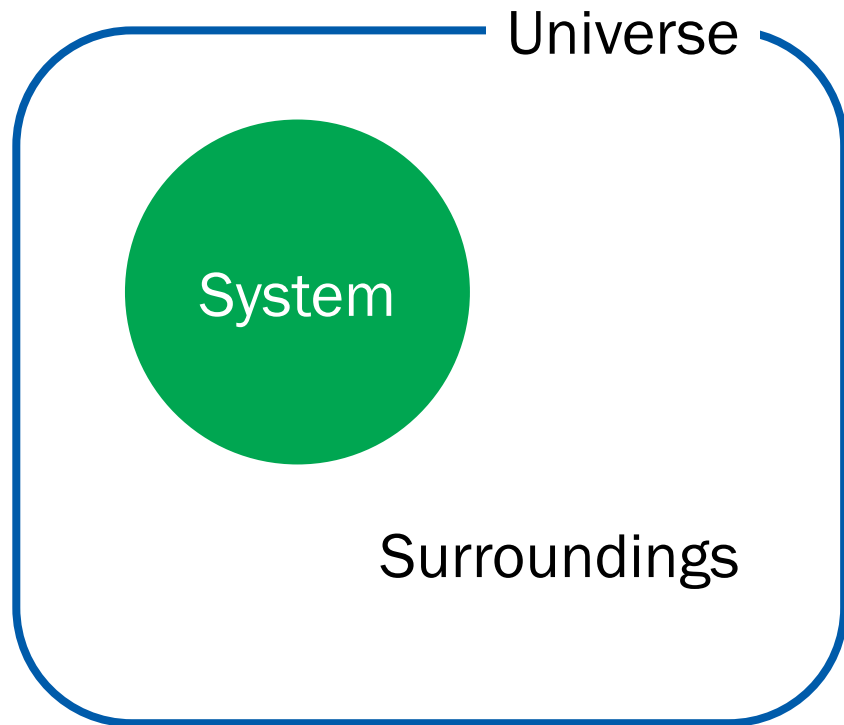
System and Surroundings



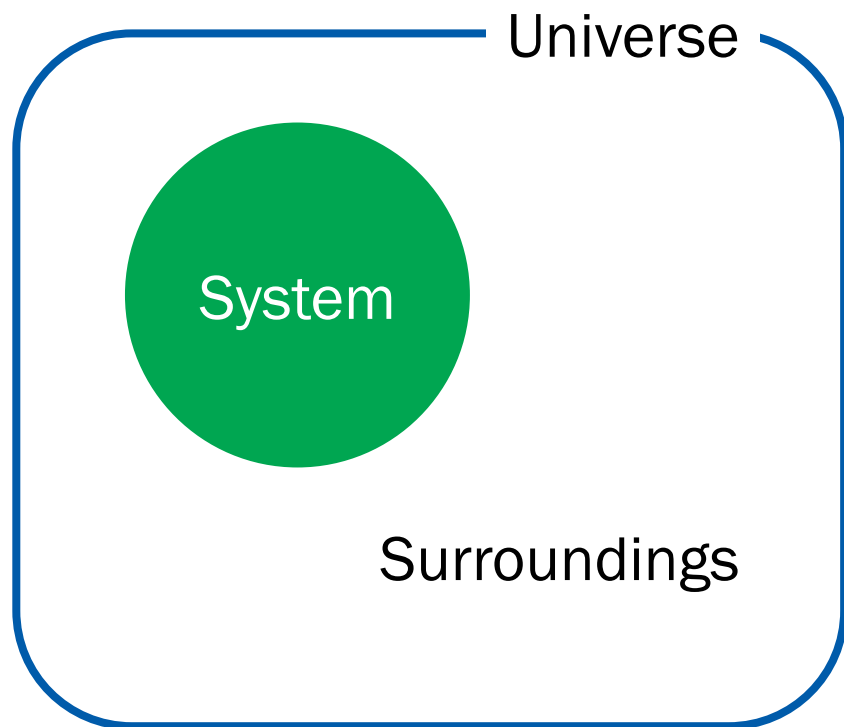
Example of a System



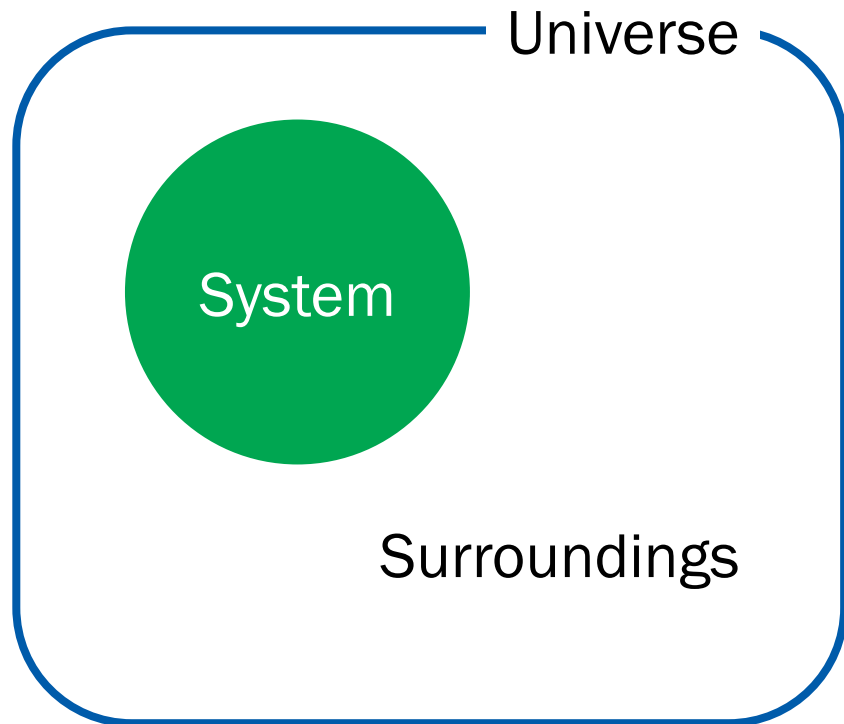
Example of a System



Another Example of a System

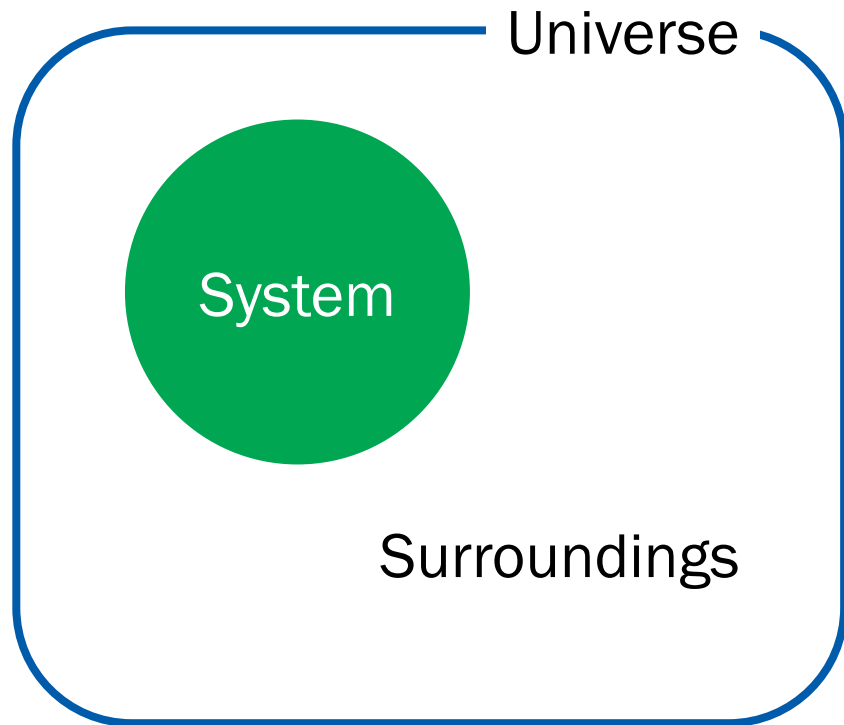


Third Example of a System

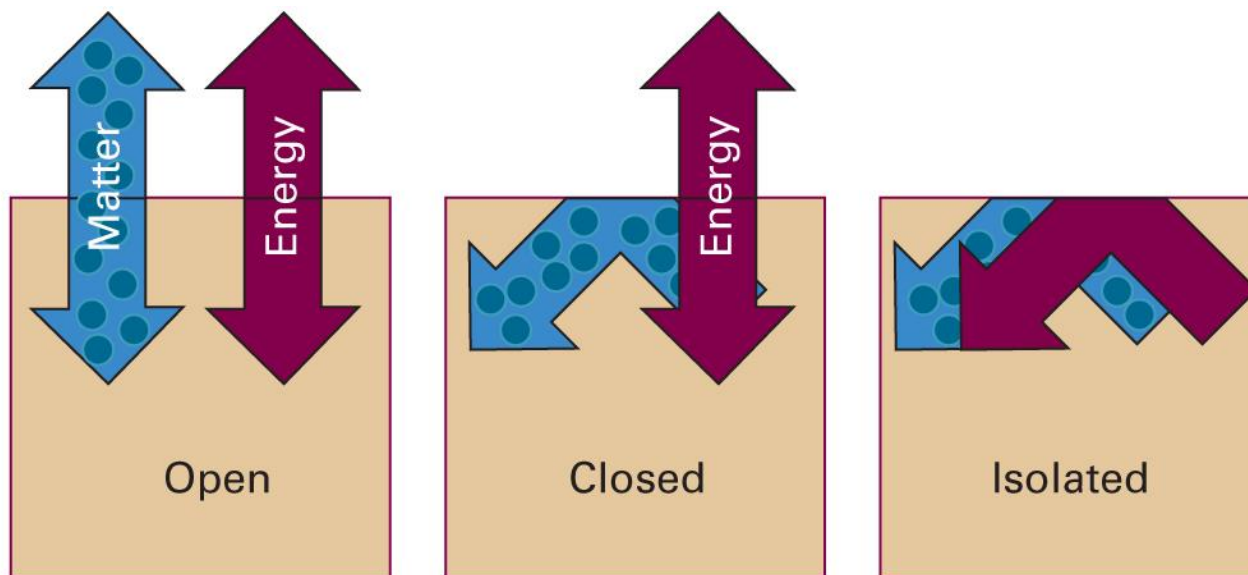


(Image: https://commons.wikimedia.org/wiki/File:Kitchen_design_at_a_store_in_NJ_refrigerator_and_cabinets_8.jpg)

And Even ...



Classification of Systems



System Properties

- A system can have different **physical properties**
- A **single value** will be assigned for each property
- Useful classification
 - **Extensive properties:** $f(A) + f(B) = f(A + B)$
 - **Intensive properties:** $f(A) + f(B) \neq f(A + B)$

Some Extensive Properties

- **Mass m :** measure of the quantity of matter present (unit: kilogram, kg)
- **Volume V :** measure of the quantity of space the sample occupies (unit: cubic meter, m³)
- **Amount of substance n :** measure of the number of specified entities (unit: mole, mol)

Relation between m and n

$$n \propto m$$

- Conversion: molar mass of a substance M (unit: kg mol^{-1})

$$n = \frac{m}{M}$$

Some Intensive Properties

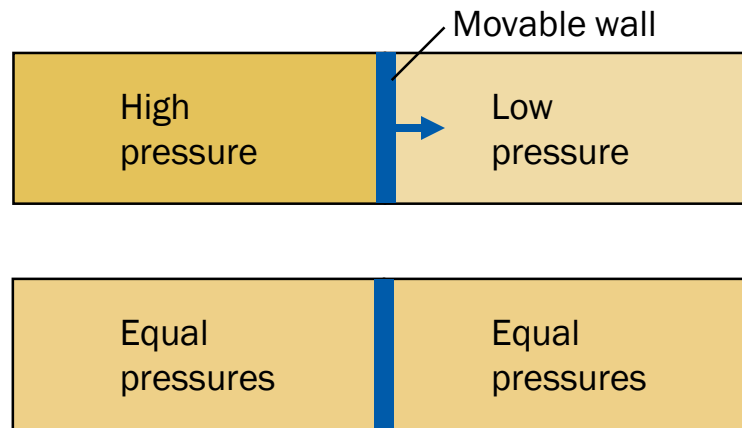
- **Pressure p :** force F / area A (unit: pascal, Pa)
 - Standard pressure $p^\ominus = 10^5 \text{ Pa} = 1 \text{ bar}$
 - Atmospheric pressure $101325 \text{ Pa} = 1 \text{ atm}$
- **Temperature T :** determines the direction of energy flow (unit: kelvin, K)
 - Celsius scale $T/\text{K} = \theta/^\circ\text{C} + 273.15$

Equilibrium

- An isolated system is **in equilibrium** when its properties remain **constant with time**
- **Equilibrium thermodynamics:** we will mostly focus on systems in equilibrium

Mechanical Equilibrium

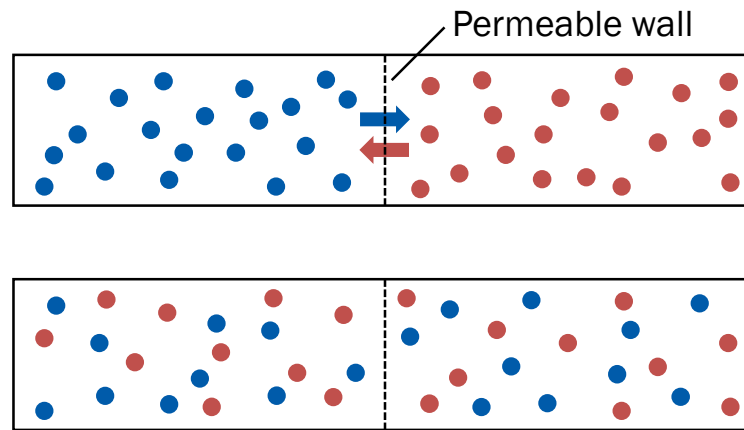
Constant p and V



Not in Atkins

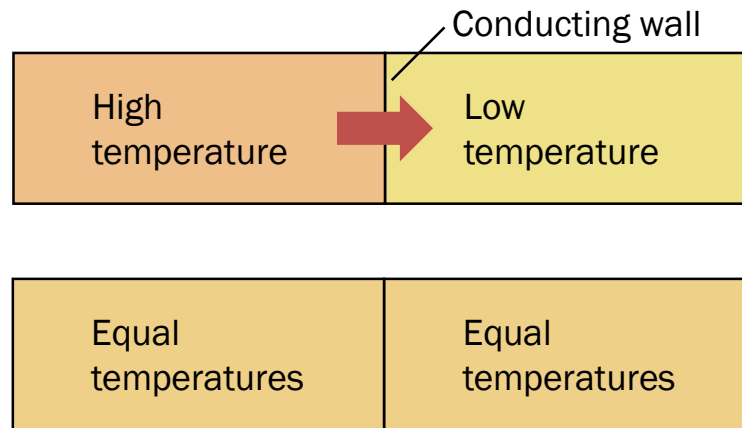
Diffusive Equilibrium

Constant n



Thermal Equilibrium

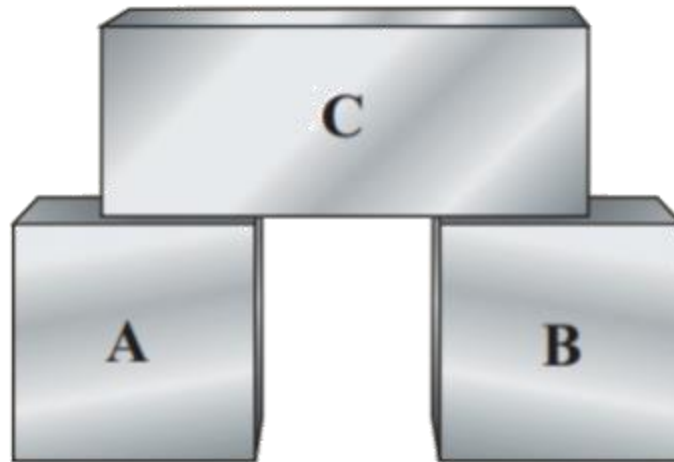
Constant T



Not in Atkins

0th Law of Thermodynamics

“If systems **A** and **B** are in thermal equilibrium, and systems **B** and **C** are in thermal equilibrium, systems **A** and **C** are in thermal equilibrium.”



(Image: http://www.brainkart.com/article/Zeroth-Law-of-Thermodynamics_36238/)

State of a System

- The **state** of a system is defined by its **properties**
- Two samples of **the same substance** that have **the same properties** are in **the same state**
- Pure substance: (n, V, p, T)

State Change (Process)

The system can change its state
from one to another

$$(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$$

Equation of State

- The state is defined by (n, V, p, T)
- **Equation of state:** relationship of the 4 variables

$$p = f(T, V, n)$$

- Depends on a system → We need a “model”

2. The Properties of Gases

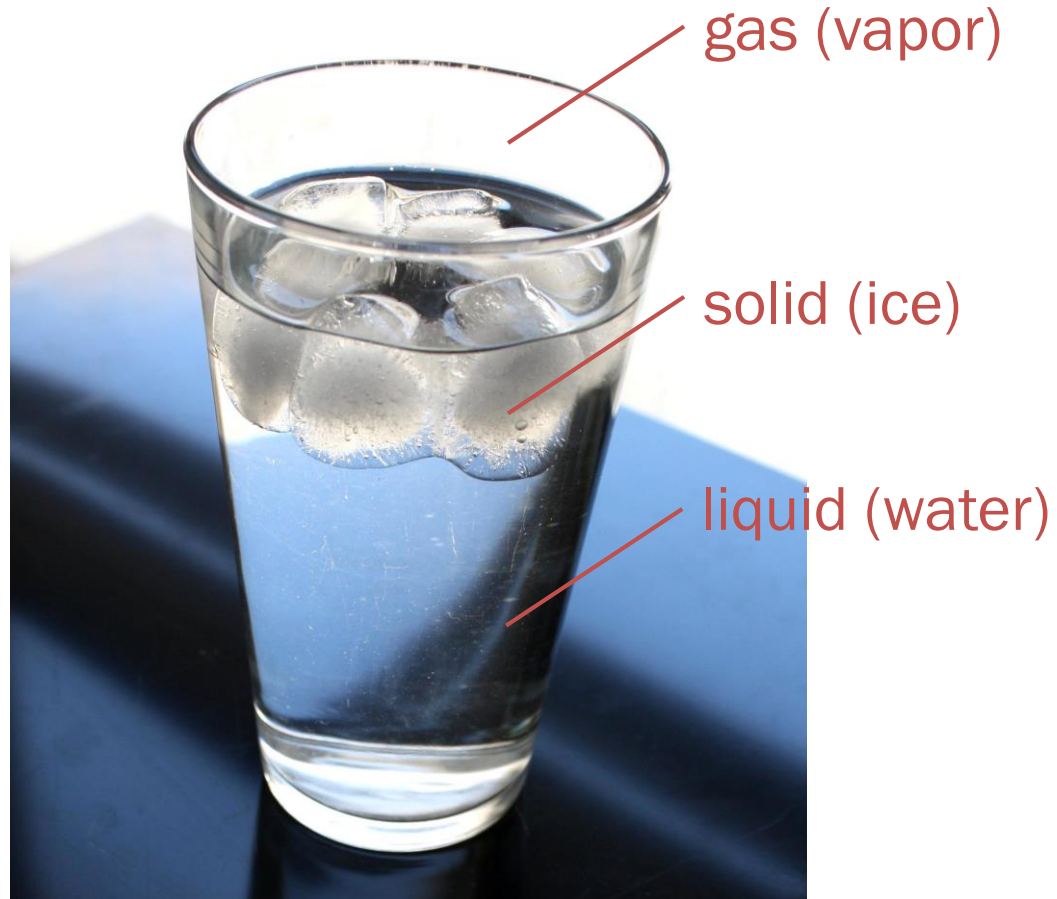


Chapter Overview

- Motivation: why gases?
- Equation of state: perfect gas
- Molecular kinetic theory
 - Maxwell-Boltzmann distribution
 - Collisions: molecule-molecule, molecule-wall
 - Kinetic energy of a gas
- Real gas

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Why Gases?

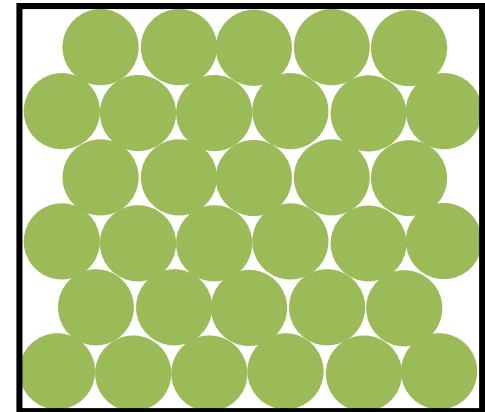
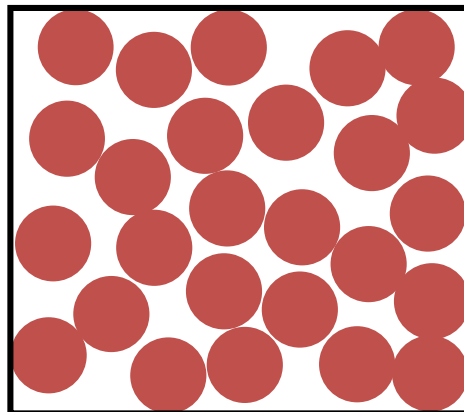
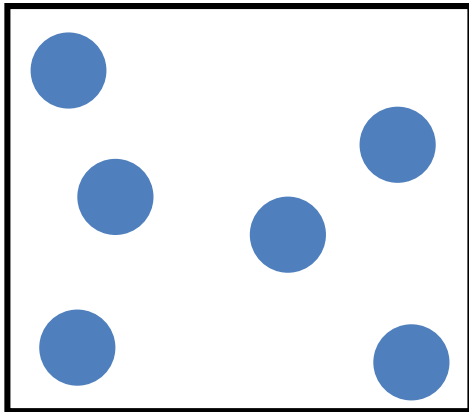


(Image: <https://pixnio.com/objects/glass/glass-ice-water>)

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Gas, Liquid, and Solid

condensed phase



Gas phase: kinetic energy $\gg$ potential energy

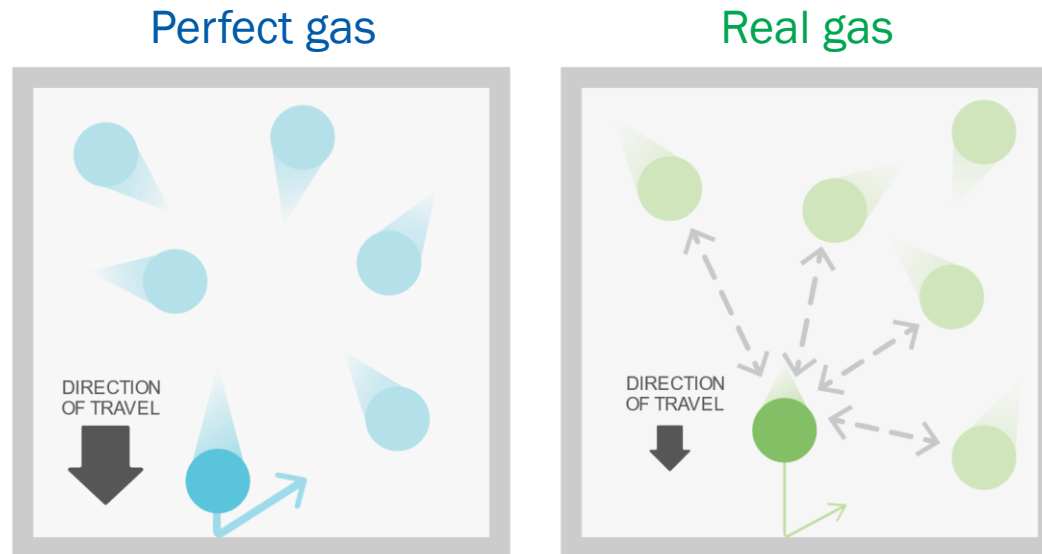
Condensed phase: kinetic energy $\approx$ potential energy

$$\frac{1}{2}mv^2$$

various forms

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



Perfect Gas Law

- **Equation of state:** relationship of the 4 variables

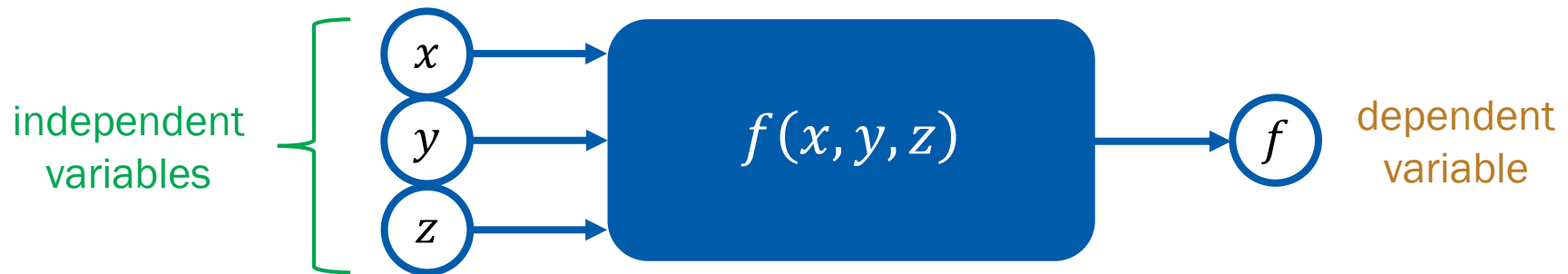
$$p = f(T, V, n)$$

- Equation of state of a perfect gas: **perfect gas law**

$$p = nRT/V$$

↑
gas constant

Multivariate Function



Example: $f(x, y, z) = x^2 + xy + 3z$

$$f(1, 1, 1) = 1 + 1 + 3 = 5$$

$$f(-1, 1, 0) = 1 - 1 + 0 = 0$$

$\vdots$

Not in Atkins

Different Forms, Same Physics

- p as a dependent variable:

$$p = f(T, V, n) = nRT/V$$

- V as a dependent variable:

$$V = f(T, p, n) = nRT/p$$

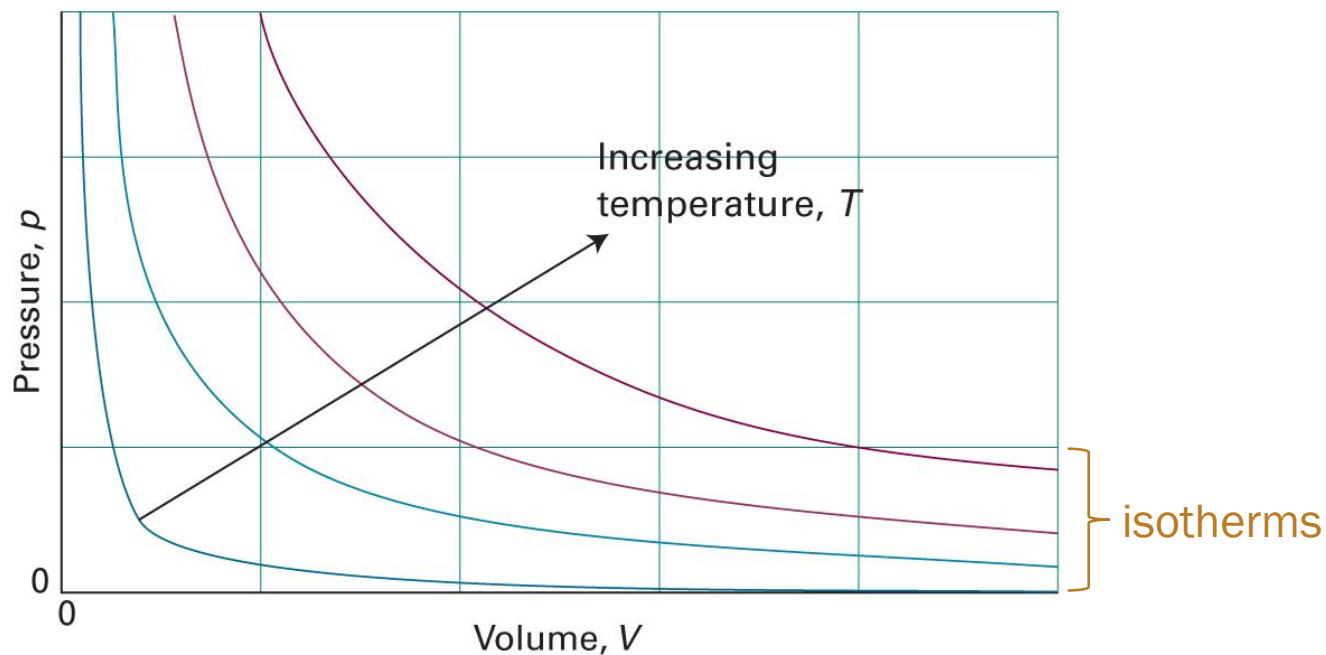
- T as a dependent variable:

$$T = f(p, V, n) = pV/nR$$



p - V Dependence

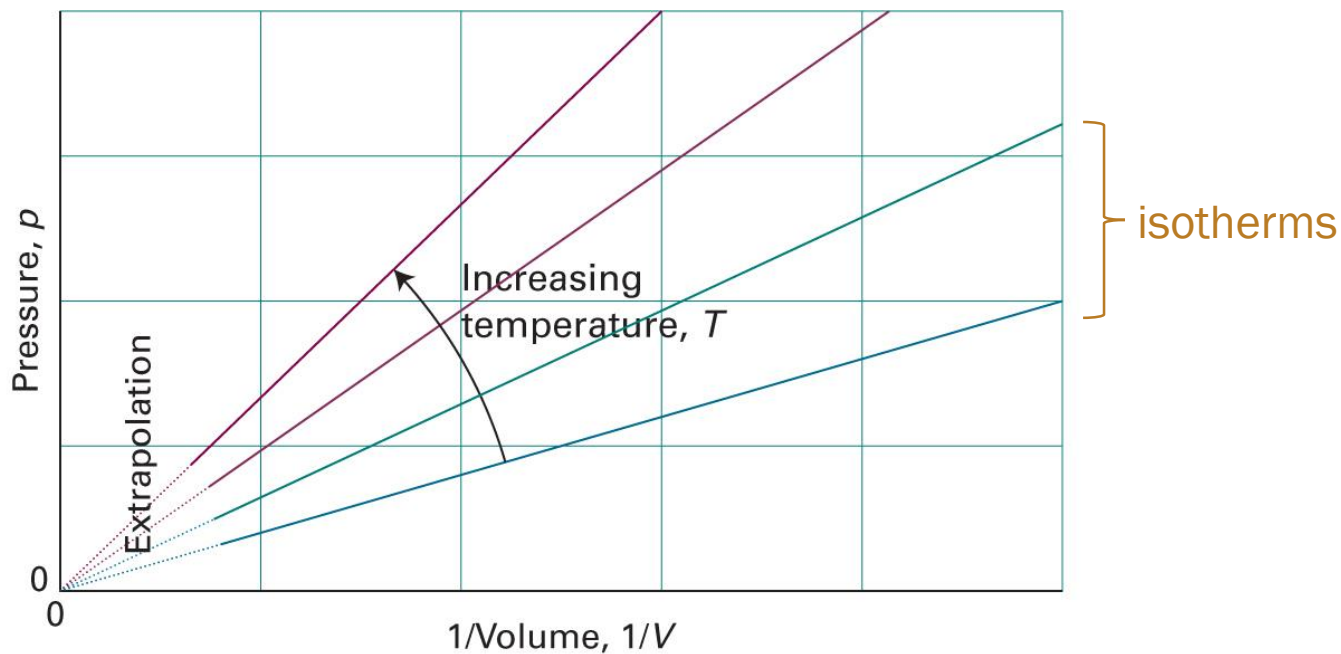
$$p = nRT/V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.2)

Another Representation

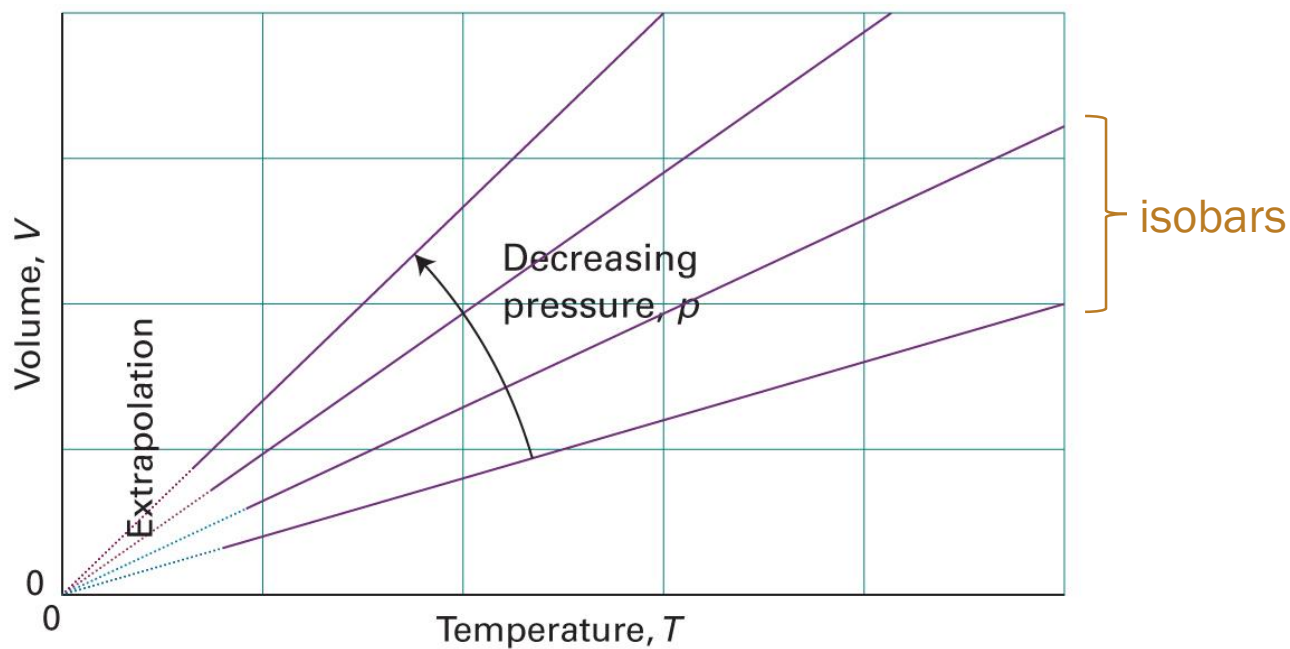
$$p = nRT/V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.3)

V - T Dependence

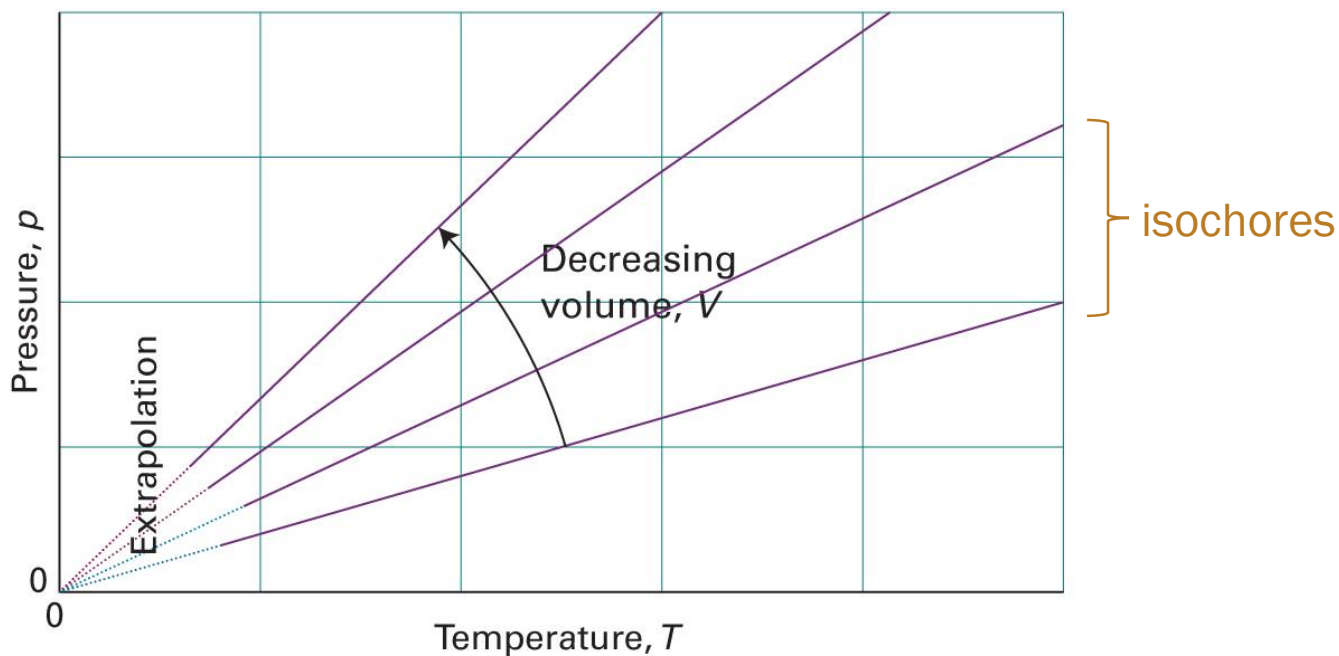
$$V = nRT/p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.4)

p - T Dependence

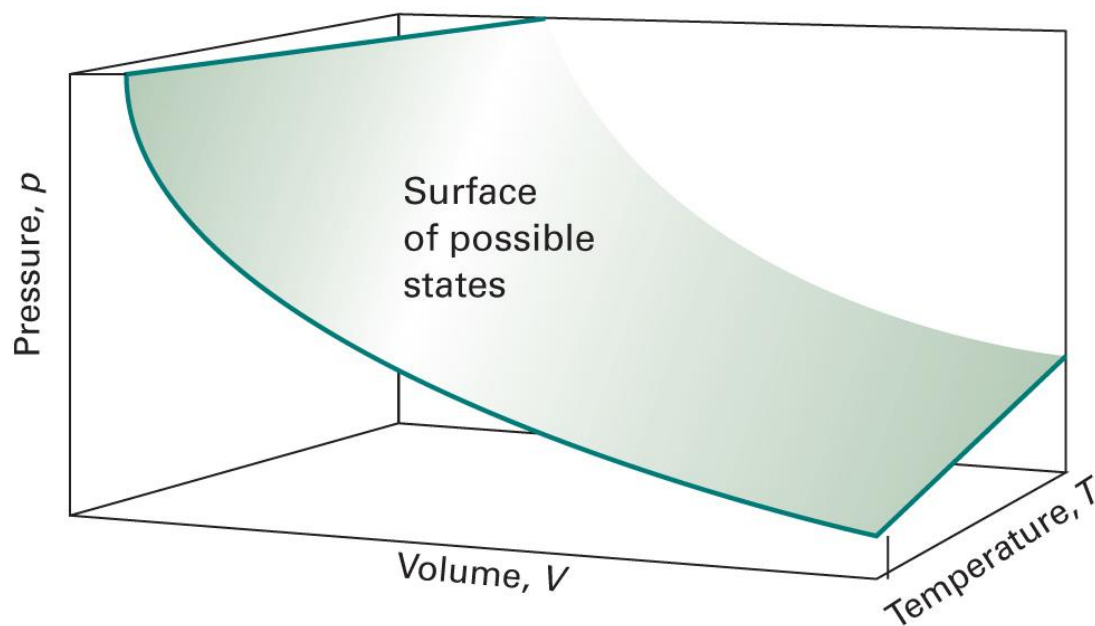
$$p = nRT/V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.5)

(p, V, T) Surface

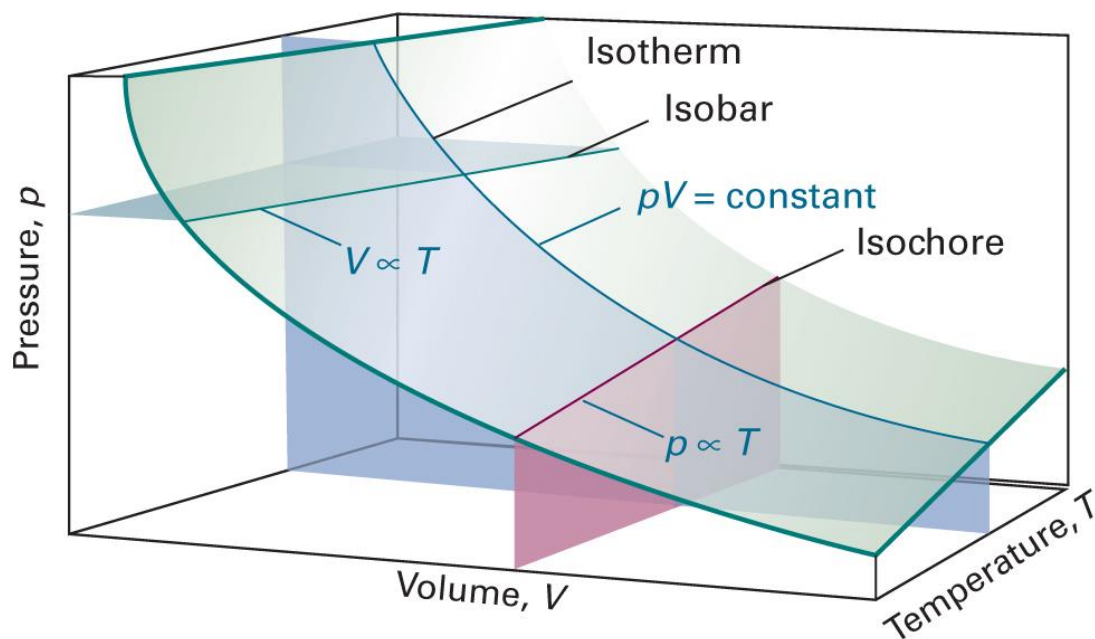
$$pV = nRT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.6)

Sections through the Surface

$$pV = nRT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1A.7)

Applications: Perfect Gas Law

- Limiting law: in the limit of $p \rightarrow 0$, all real gases behave as a perfect gas
- STP (standard temperature and pressure)
 - $p = 1 \text{ atm}$, $T = 0^\circ \text{C} = 273.15 \text{ K}$
 - $V/n = 22.414 \text{ L mol}^{-1}$
- SATP (standard ambient temperature and pressure)
 - $p = 10^5 \text{ Pa}$, $T = 25^\circ \text{C} = 298.15 \text{ K}$
 - $V/n = 24.789 \text{ L mol}^{-1}$

Combined Gas Law

From the perfect gas law,

$$R = \frac{pV}{nT} = \text{constant}$$

Hence, for a change from state 1 to state 2,

$$\frac{p_1 V_1}{n_1 T_1} = \frac{p_2 V_2}{n_2 T_2}$$

Can we explain the behavior of perfect gas?

Kinetic Molecular Theory

Total energy = Σ (kinetic energies of molecules)

Speed distribution

Collision frequency

Pressure

Internal Energy

⋮

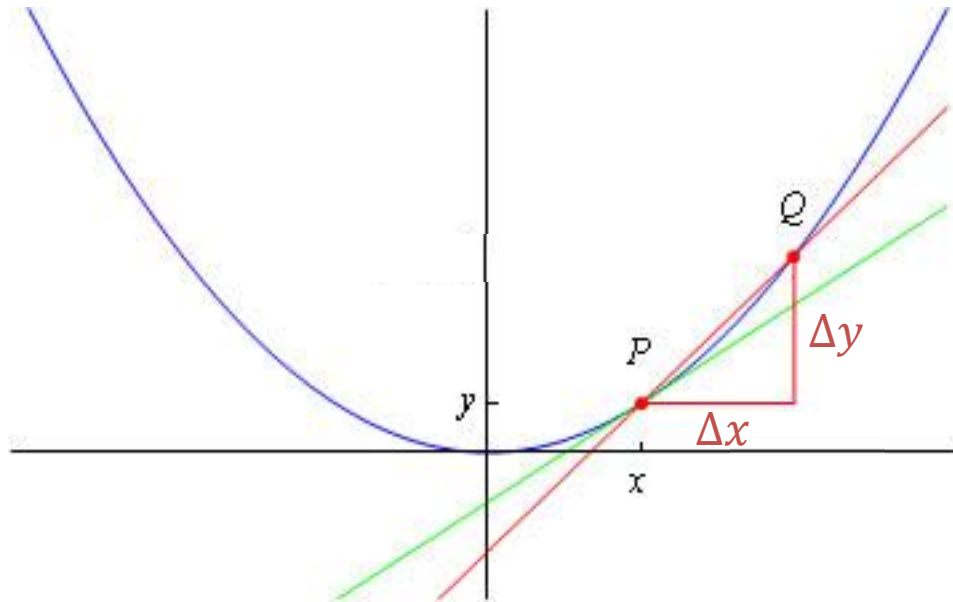
Fundamental Assumptions

1. Molecules with mass m in ceaseless random motion
2. Molecules with negligible size
3. Elastic collisions: the total kinetic energy is conserved

Described by classical mechanics

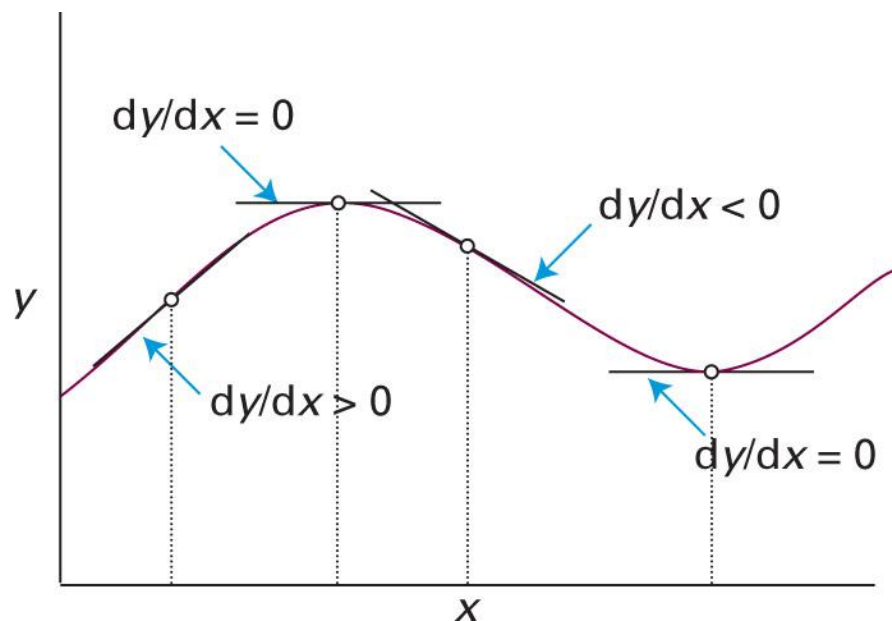
Differentiation

“The slope of a function”



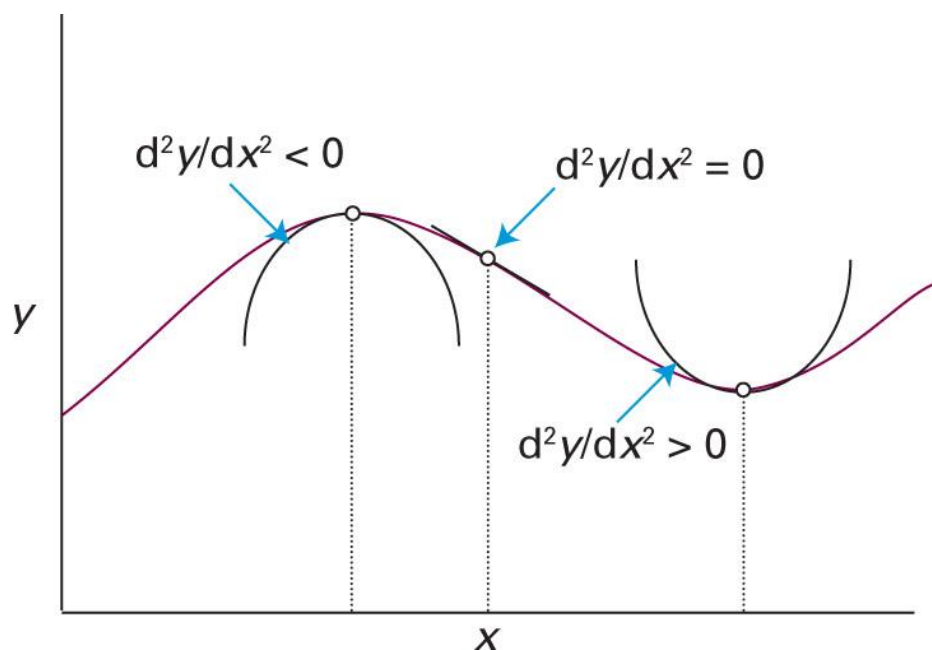
$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

First Derivative: Slope



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., the chemist's toolkit 5, Sketch 1)

Second Derivative: Sharpness



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., the chemist's toolkit 5, Sketch 2)

Derivatives of Some Functions

$$\frac{d}{dx} a = 0$$

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^{ax} = ae^{ax}$$

$$\frac{d}{dx} \ln ax = \frac{1}{x}$$

$$\frac{d}{dx} \sin ax = a \cos ax$$

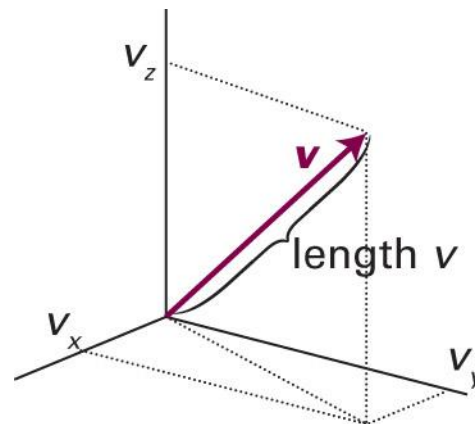
$$\frac{d}{dx} \cos ax = -a \sin ax$$

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

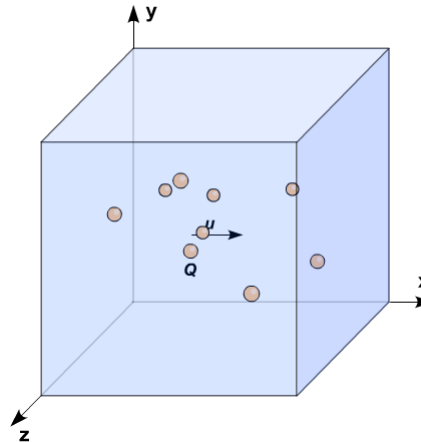


With the velocity, we can define:

- Speed $v = |\vec{v}| = (v_x^2 + v_y^2 + v_z^2)^{1/2}$
- Kinetic energy $E_k = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Molecular Velocities

Each molecule has a different velocity



How can we determine the information about velocity?